

Aggregation and EXTEND

txt2tex example

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[*InvoiceID*, *PaymentID*, *Currency*]

Payment

paymentId : *PaymentID*

invoiceId : *InvoiceID*

amountPaid : *Currency*

Group aggregate: one summary row per group

InvoicePayments == Project_{*invoiceId*,*paymentId*}(*Payment*)

InvoicePayments Group(Count(*paymentId*) as *numPayments*)

Nest then EXTEND: a multiplicity-preserving sum

InvPay == Project_{*invoiceId*,*paymentId*,*amountPaid*}(*Payment*)

InvPayNested == *InvPay* GROUP({*paymentId*, *amountPaid*} AS *payments*)

PaidPerInv == *InvPayNested* Extend(Sum(*payments*, *amountPaid*) as *totalInvoicePayment*)

InvPayNested Extend(Sum(*payments*, *amountPaid*) as *totalPaid*, Count(*payments*) as *numPaid*)

Cartesian product

InvoicePayments × *InvoicePayments*